RotateAttention: RoPE-Aware Rotation and Range Rectification for INT4 Quantized Attention in Video Generation

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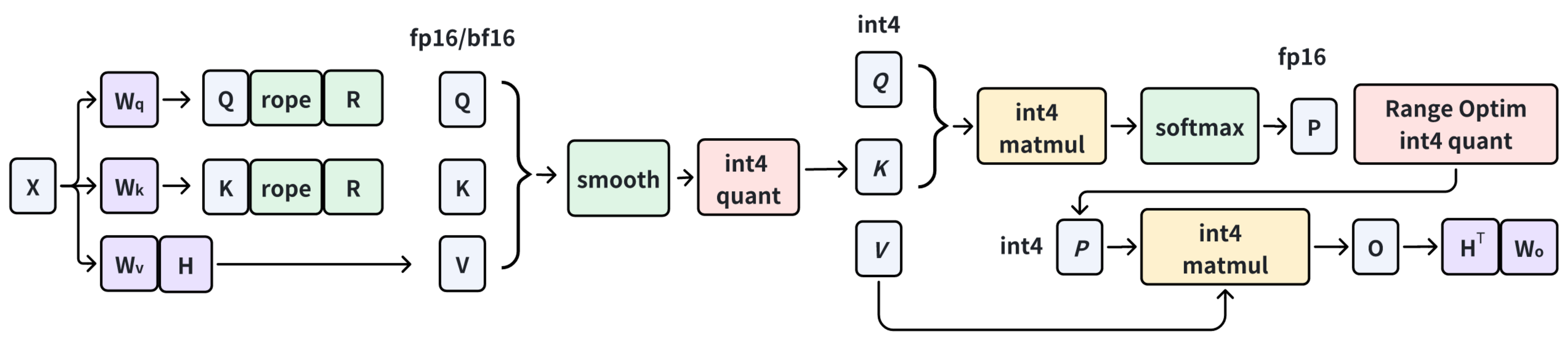
Problem Definition & Contribution

Goal: Fast, accurate **INT4 quantized FlashAttention** for **3D-RoPE video DiTs** (sequences $>$ 10K tokens, quadratic cost).

- **C1:** online rotation (the usual Q/K outlier remedy) is hard to reconcile with **3D RoPE**.
- **C2:** $\mathbf{P} \in (0, 1]$ makes symmetric INT4 waste *half* of the range.
- **Observation:** 3D RoPE induces structured, *segmental* Q/K outliers with cross-matrix symmetry.

Contributions:

- **RoPE-aware Rotation** (Interleaved/Half) equalizes 3D-RoPE outliers at zero/negligible cost.
- **Key finding: rotation matrices must be orthogonal**; unconstrained rotations amplify quantization error.
- **Range-optimized $\mathbf{P}$** and **Hadamard $\mathbf{V}$** (zero cost) complete the INT4 pipeline.
- Up to **2.2**$\times$ kernel / **1.68**$\times$ end-to-end speedup, near-FP16 quality.

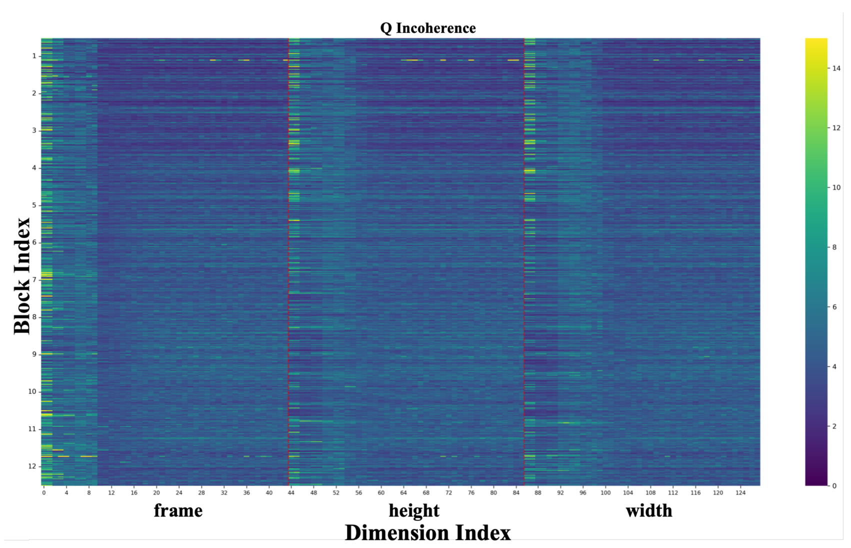


Overall pipeline: RoPE-aware rotation, range-optimized $\mathbf{P}$ quantization, and Hadamard $\mathbf{V}$ transform.

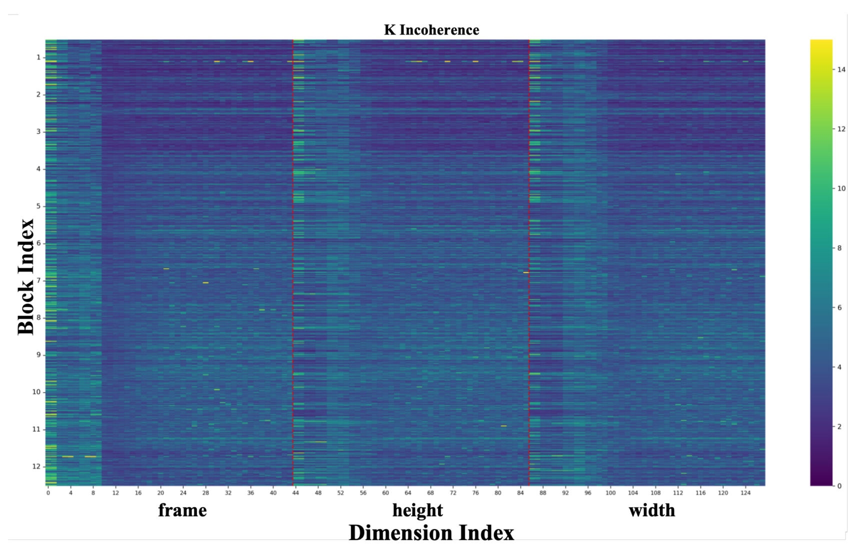
Outlier Analysis: What 3D RoPE Does to Q/K

Metric: $\Psi(\mathbf{X}) = \max_i \left|\mathbf{X}_{i,:} - \mathbb{E}[\mathbf{X}]\right|$ (channel-level incoherence); 3D RoPE partitions $D = D_f + D_h + D_w$.

- **Segmental incoherence:** outliers split into frame/height/width segments; one half of each segment dominates.
- **Cross-matrix symmetry:** $\mathbf{Q}$ and $\mathbf{K}$ share nearly identical incoherence.



$\mathbf{Q}$ (Wan2.2)



$\mathbf{K}$ (Wan2.2)

Take-away: structured, symmetric patterns make *orthogonal, RoPE-compatible* rotations the right tool.

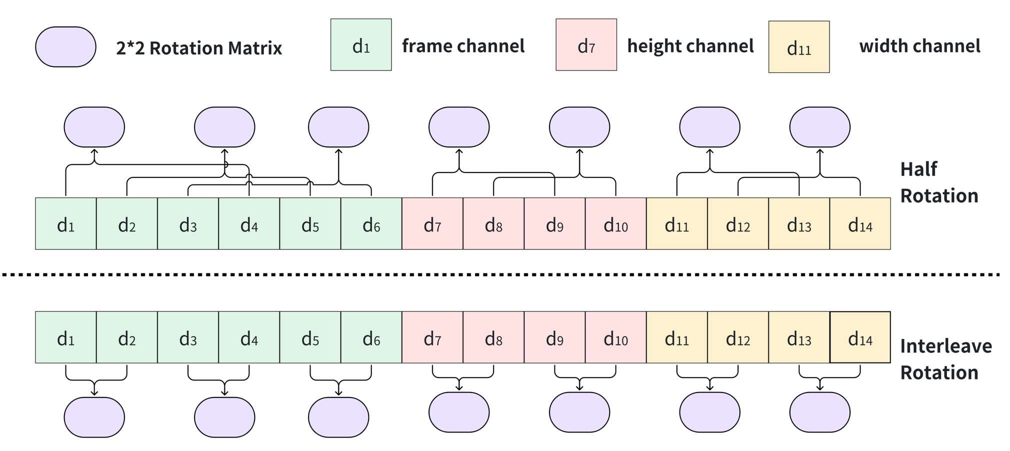
Method: RoPE-aware Rotation + Range-optimized P

(1) RoPE-aware Rotation. Block-diagonal orthogonal $\mathbf{R} = \mathrm{diag}(\{\mathbf{R}_i\})$, $\mathbf{R}_i \in \mathrm{O}(2)$.

- **Interleaved:** $\mathrm{RoPE}(\mathbf{Q})\mathbf{R} = \mathbf{Q}(\mathbf{MR})$ pre-fused (zero cost).
- **Half:** couples d_i^j with $d_{i+D_j/2}^j$ within segment $j \in \{f, h, w\}$.

(2) Range-optimized P: $\tilde{\mathbf{P}} = \lfloor \mathbf{P}/\max(\mathbf{P}) \times 15 \rceil - 8 \in (-8, 7]$ (full 16 levels).

(3) Hadamard V: merge $\widetilde{\mathbf{W}}_v = \mathbf{W}_v\mathbf{H}$, $\widetilde{\mathbf{W}}_o = \mathbf{H}^\top\mathbf{W}_o$ offline.



Qualitative Results (Generated Video Frames)

Wan2.2 T2V (top) and Wan2.2 I2V (bottom). Our rotations with range-optimized $\mathbf{P}$ keep generated frames close to FP16 and clearly improve structure and temporal consistency over SageAttention.

Learned Rotation Must Be Orthogonal

Learned rotations $\in \mathrm{O}(2)$ (orthogonal constraint).
Limited rotation space: no clear quality gain, but no degradation.
Without the constraint $\rightarrow$ degraded results.